Tutorial on Implicit Neural Representations for Medical Imaging

TUM HARVARD UNIVERSITY MICCAI 2024 Marrakesh MOROCCO UNIVERSITY OF TWENTE. MRI TUM mcml

Tutorial on
Implicit Neural Representations for Medical Imaging

Tutorial Schedule

Introduction (08.00 - 08.40) (Julian):

- *What are INRs and Neural Fields?*
- *What do we need to train INRs - and why would we want to train them in the first place?*
- *Formal Definitions & Intuitions for Medical Imaging*

Practical Coding (8.40 - 09.20) (Nil):

- *It's easy! How to train your first INR - and what's different in comparison to classical deep learning applications*
- *Using some hacks to make INRs work*
- *Demoing a real-world INR application in k-space*

In-depth Guest Lecture (9.20 - 10.00) (Jelmer):

- *Applications of INRs in registration*
- *Applications of INRs in shape modeling*

Coffee Break (10.00 - 10.30)

Overview of INR applications and domains (10.30-11.00) (Veronika, Supro) :

- *Applications in Undersampling, Reconstruction and Rendering*
- *Applications in Segmentation, Registration*
- *Applications in Temporal Dynamics and Progression Modeling*

Guest Lecture on Advanced Concepts of INRs (11.00-11.50) (Jonathan):

- *Cohort-based Learning and Meta Learning for Initialization*
- *INRs for Compression and From Data to Functa*

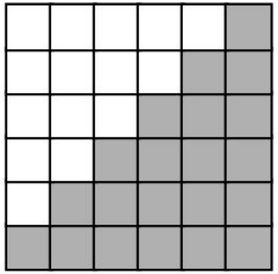
Cohort Learning, Neural Rendering & Conclusion (11.50 - 12.30) (Maik, Vasiliki):

- *An application of cohort-based learning in fetal imaging*
- *Neural Rendering and frameworks to start your research*
- *Future Research Directions & Conclusion*

Source: Scaniverse by Niantic, https://www.youtube.com/watch?v=1LQh_Im0VoM

What are Implicit Neural Representations?
What are Neural Fields?
And how did the journey begin?

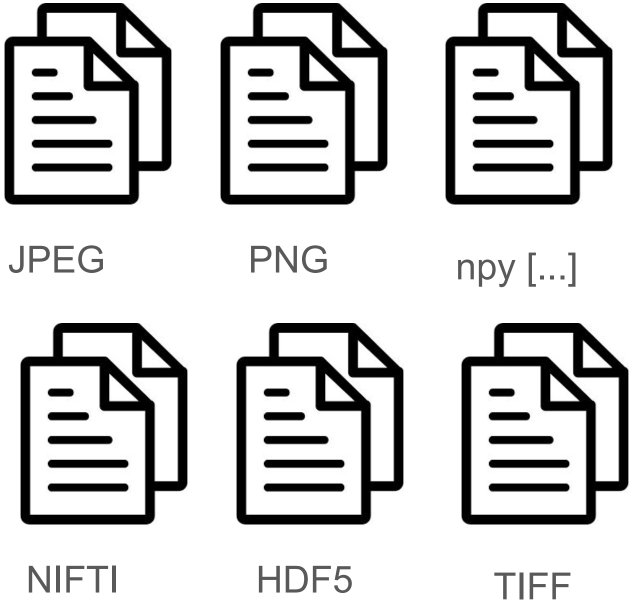
Preface - how it all began …



Voxels

[0, 0] [0]
[0, 1] [1]
[…]

2D: $(x, y) \rightarrow y \in [0, 1]$



Mescheder L, Oechsle M, Niemeyer M, Nowozin S, Geiger A. Occupancy Networks, CVPR 2019
Park JJ, Florence P, Straub J, Newcombe R, Lovegrove S. DeepSDF, CVPR 2019

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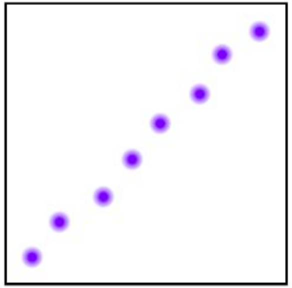
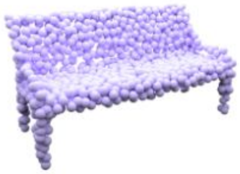




Point Clouds

[0.25, 0.12] [1]
[0.33, 0.11] [0]
[...]

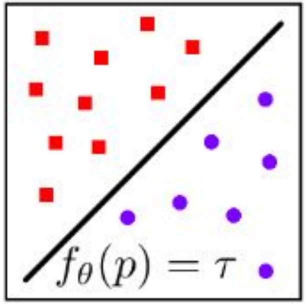
2D: $(x, y) \rightarrow y \in [0, 1]$

PLY XYZ npy

Mescheder L, Oechsle M, Niemeyer M, Nowozin S, Geiger A. Occupancy Networks, CVPR 2019
Park JJ, Florence P, Straub J, Newcombe R, Lovegrove S. DeepSDF, CVPR 2019

Is there a more compact and accurate way of representing an object?

Preface - how it all began …



Instead of storing individual points or meshes, can we store the underlying function of the decision boundary?

Mescheder L, Oechsle M, Niemeyer M, Nowozin S, Geiger A. Occupancy Networks, CVPR 2019
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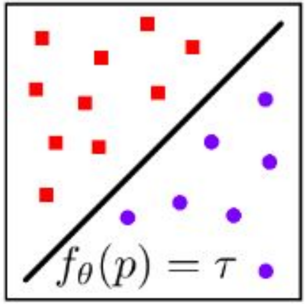

Instead of storing individual points or meshes, can we store the underlying function of the decision boundary?

But we do not know the function - or do we?

Mescheder L, Oechsle M, Niemeyer M, Nowozin S, Geiger A. Occupancy Networks, CVPR 2019
Park JJ, Florence P, Straub J, Newcombe R, Lovegrove S. DeepSDF, CVPR 2019

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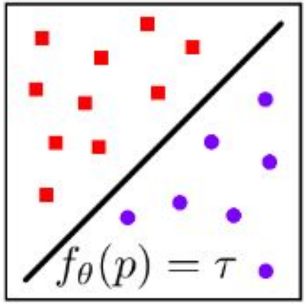

We can learn this function from observations:

(1) We overfit a model given some (sparse) observations using a neural network
(2) We learn a continuous function since the input domain is continuous.
(3) To visualize the object, we no longer need the points - we can query the network and ask for the signal value.

Mescheder L, Oechsle M, Niemeyer M, Nowozin S, Geiger A. Occupancy Networks, CVPR 2019
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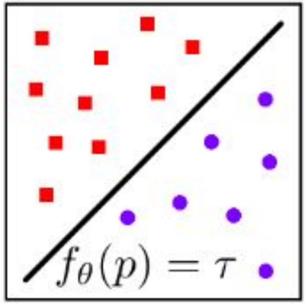

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→ we now have an implicit neural representation

Implicit Neural Representation

Mescheder L, Oechsle M, Niemeyer M, Nowozin S, Geiger A. Occupancy Networks, CVPR 2019
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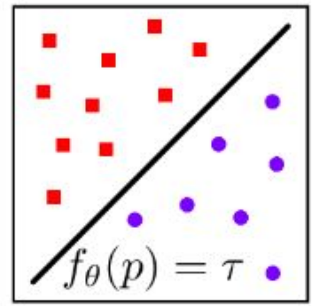
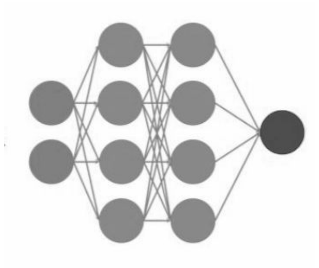
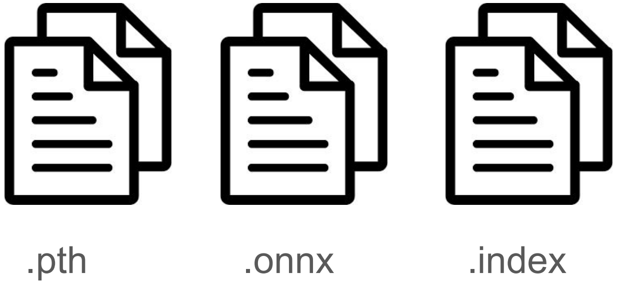


Implicit Neural Representation

Finally, how does our representation look like?





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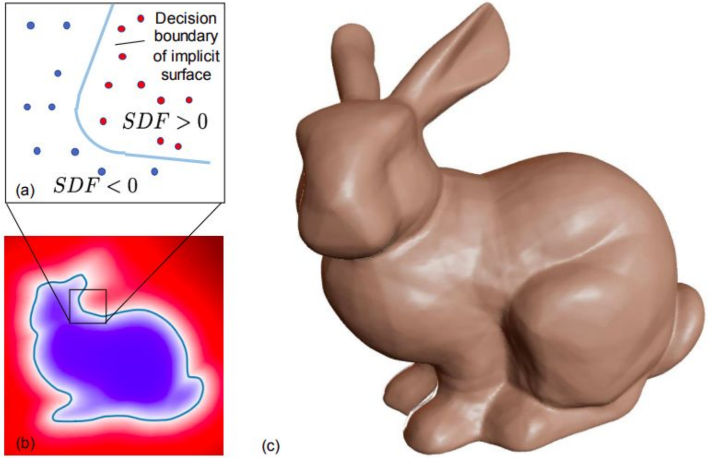
Why is this interesting?

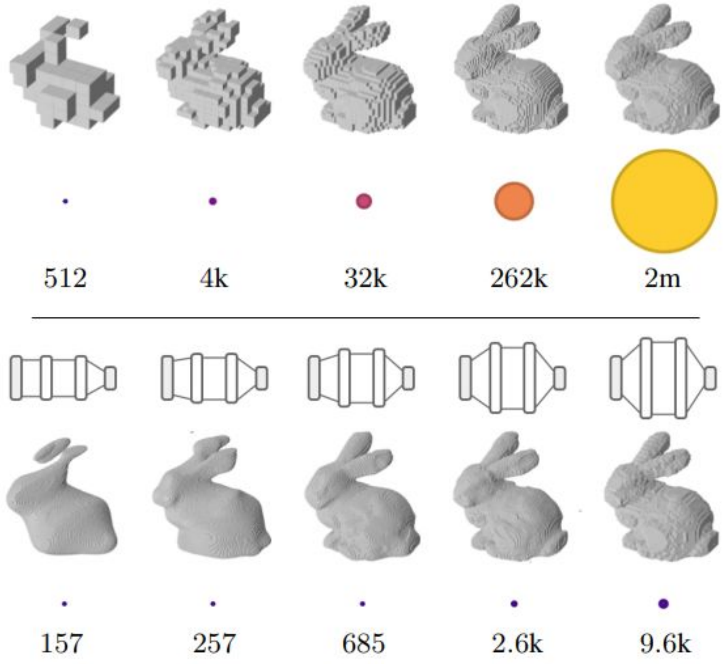
Advantages of implicit neural representations



INRs allow us to more accurately represent the underlying data that does not live on a grid.

Park JJ, Florence P, Straub J, Newcombe R, Lovegrove S. DeepSDF, CVPR 2019

Advantages of implicit neural representations



While traditional data representations as grids and arrays scale with resolution, neural networks scale with complexity.

Dupont, Emilien, et al. "COIN++: Neural Compression Across Modalities." *TMLR 2022*

Are implicit neural representations limited to shape data only?

Applications of implicit representations in CV

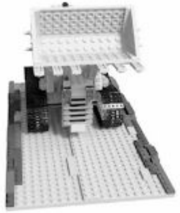
Color Image

$$f_\theta(x, y) = (r, g, b)$$

Audio Signal / Time Series Data

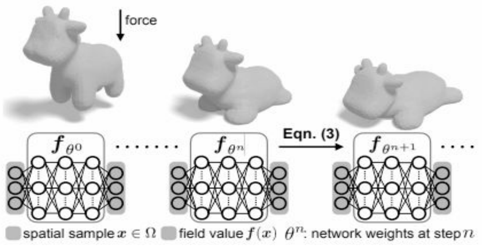
$$f_\theta(t) = a$$

Neural Radiance Field (NeRF)



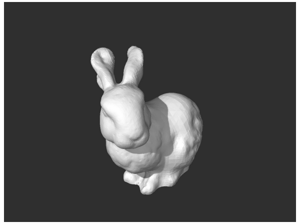
$$f_\theta(x, y, z, d) = (\sigma, r, g, b)$$

(Partial) Differential Equations



$$f_\theta(x, y, t) = u$$

Signed Distance Function



$$f_\theta(x, y, z) = d$$

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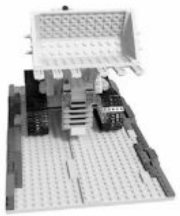

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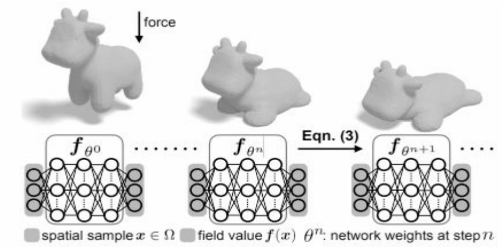

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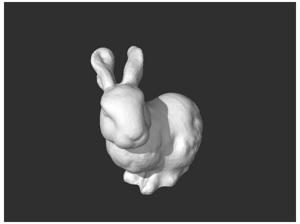

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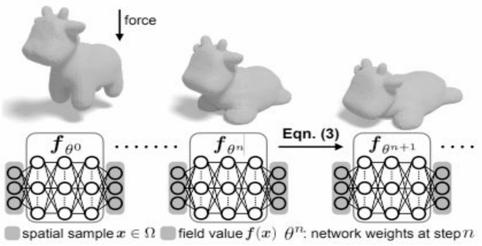

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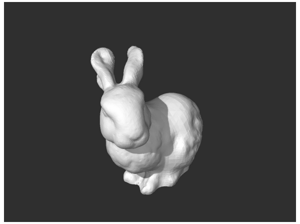

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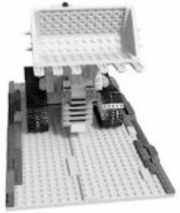

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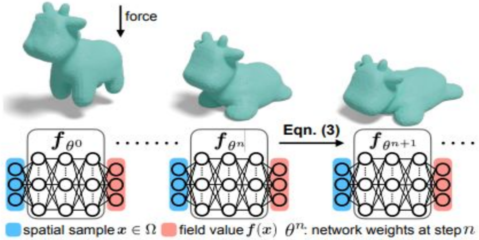

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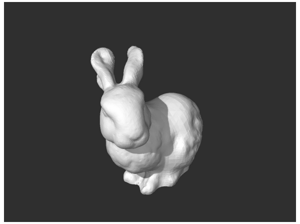

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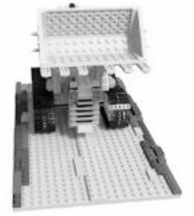

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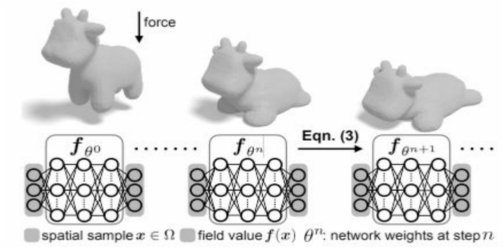

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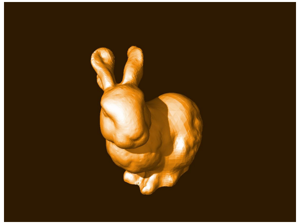


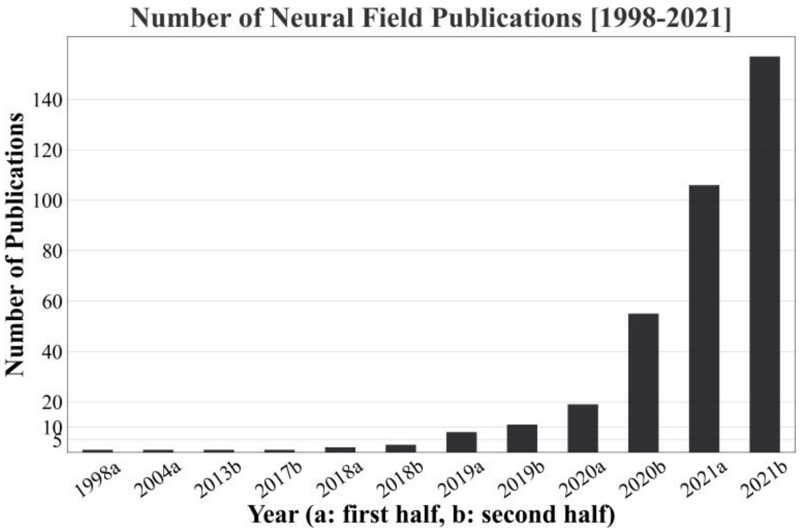
$$f_\theta(x, y, z) = d$$

...

Let’s properly define an INR! :)

INRs - Terminology and related fields

«Within visual computing, neural fields have been called implicit neural representations, neural implicits, or coordinate-based neural networks. »



Xie Y, Takikawa T, Saito S, Litany O, Yan S, Khan N, Tombari F, Tompkin J, Sitzmann V, Sridhar S. Neural fields in visual computing and beyond. In: Computer Graphics Forum 2022 May (Vol. 41, No. 2, pp. 641-676).

Defining implicit representations

Definition 1: A *field* is a quantity defined for all *spatial and/or temporal coordinates*.

Feynman R. P., Leighton R. B., Sands M.: The Feynman Lectures on Physics; Vol. I. American Journal of Physics 33, 9 (1965), 750–752.
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Defining implicit representations

Definition 1: A *field* is a quantity defined for all spatial and/or temporal coordinates.

Definition 2: A *neural field* is a field that is parameterized fully or in part by a *neural network*.

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$$s = f_\theta(x)$$

- spatio-temporal coordinates $x \in R^n$
- signal value $s \in R^m$ at a spatio-temporal coordinate x (e.g., pixel color, sound pressure level, or occupancy value)
- the continous function $f_\theta : R^n \to R^m$ that maps the input x to the output s
- θ represents the set of parameters (weights) of the neural network that are learned during training

Feynman R. P., Leighton R. B., Sands M.: The Feynman Lectures on Physics; Vol. I. American Journal of Physics 33, 9 (1965), 750–752.
Xie Y, Takikawa T, Saito S, Litany O, Yan S, Khan N, Tombari F, Tompkin J, Sitzmann V, Sridhar S. Neural fields in visual computing and beyond. In: Computer Graphics Forum 2022 May (Vol. 41, No. 2, pp. 641-676).

Adjacent fields - Physics Informed Neural Networks

Physics Informed Neural Networks:

«Finally, neural fields are domain-agnostic and can model arbitrary quantities beyond shape and appearance. Neural fields have been used extensively in computational physics *via physics-informed neural networks (PINNs).*»

«*Physics-informed neural networks (PINNs)* use learning bias which supervises boundary and initial values (from an incomplete simulation or observations) using a loss, and the rest of space by *sampling or regularizing with equations of physics, typically partial differential equations (PDEs).* »

Xie Y, Takikawa T, Saito S, Litany O, Yan S, Khan N, Tombari F, Tompkin J, Sitzmann V, Sridhar S. Neural fields in visual computing and beyond. In: Computer Graphics Forum 2022 May (Vol. 41, No. 2, pp. 641-676).

Adjacent fields - Physics Informed Neural Networks

MICCAI 2024 Marrakesh MOROCCO

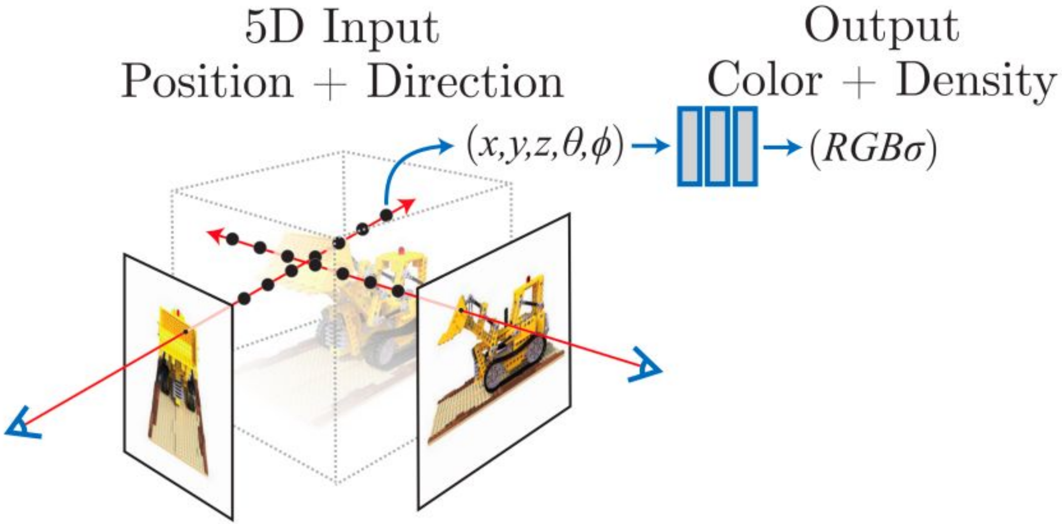
October 6	Data Learning meets Computational Modelling: Successfully using Physics-Informed Neural Networks for Biomedical Applications	PINNs	

https://annien094.github.io/PINNs-tutorial-MICCAI-2024/

Last but not least - what are NeRFs?

Neural Radiance Fields (NeRF)

NeRF constitutes a neural field that *maps coordinates (position+direction)* to *radiance and density.* Typically, it is posed as an inverse rendering problem given sparse 2D observations.



Mildenhall, Srinivasan, Tancik et al., NeRF: Representing scenes as neural radiance fields for view synthesis, *ECCV'22*

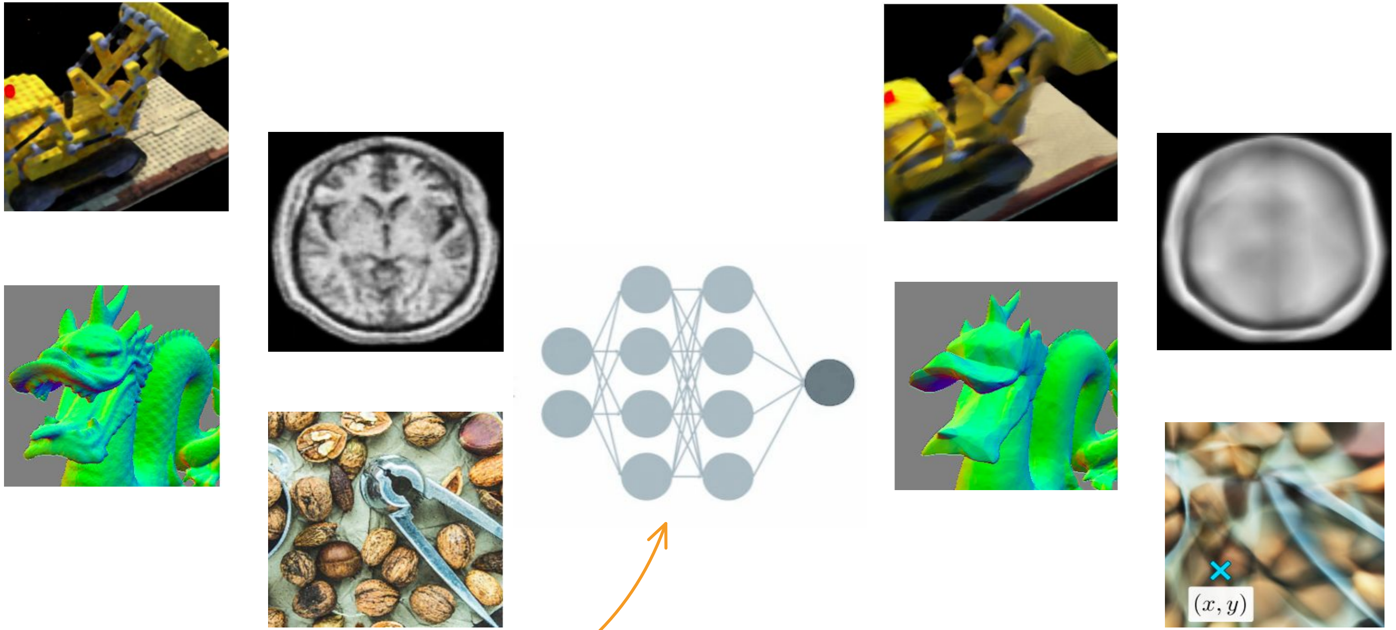
How do we train INRs?
(Hands-on in the next session)

INRs - training process

Tancik M, Srinivasan P, Mildenhall B, Fridovich-Keil S, Raghavan N, Singhal U, Ramamoorthi R, Barron J, Ng R. Fourier features let networks learn high frequency functions in low dimensional domains. Advances in neural information processing systems. 2020;33:7537-47.
Video: https://www.youtube.com/watch?v=nVA6K6Sn2S4&t=11s

A simple MLP - we train it & we are done?

Mitigating the low-frequency bias

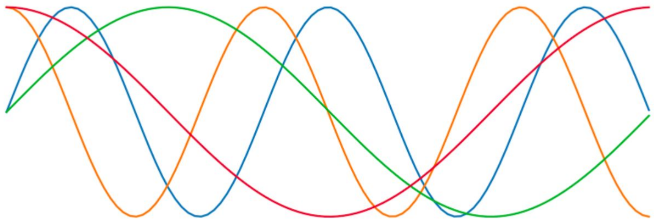


→ *ReLU-MLPs* have a bias towards learning low frequencies

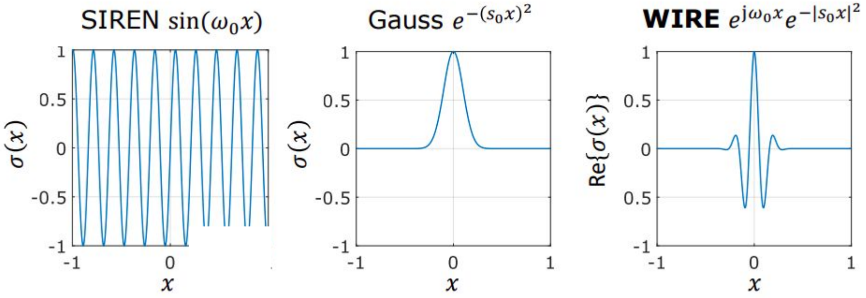
Tancik M, Srinivasan P, Mildenhall B, Fridovich-Keil S, Raghavan N, Singhal U, Ramamoorthi R, Barron J, Ng R. Fourier features let networks learn high frequency functions in low dimensional domains. Advances in neural information processing systems. 2020;33:7537-47.

Mitigating the low-frequency bias

Coordinate Encoding / Projection



Non-ReLU Activation Functions



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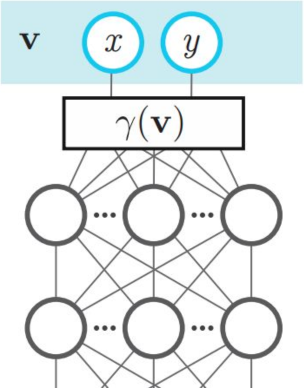
Saragadam V, LeJeune D, Tan J, Balakrishnan G, Veeraraghavan A, Baraniuk RG. Wire: Wavelet implicit neural representations. InProceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition 2023 (pp. 18507-18516).

Mitigating the low-frequency bias using Fourier Features

Fourier Feature Mapping:

$$\gamma(\mathbf{v}) = \left[a_1 \cos(2\pi \mathbf{b}_1^{\mathrm{T}} \mathbf{v}), a_1 \sin(2\pi \mathbf{b}_1^{\mathrm{T}} \mathbf{v}), \ldots, a_m \cos(2\pi \mathbf{b}_m^{\mathrm{T}} \mathbf{v}), a_m \sin(2\pi \mathbf{b}_m^{\mathrm{T}} \mathbf{v})\right]^{\mathrm{T}}$$



→ a simple Fourier Feature encoding enables the MLP to learn high frequencies

Tancik, Matthew, et al. "Fourier features let networks learn high frequency functions in low dimensional domains." *NIPS 2020*

Mitigating the low-frequency bias using Fourier Features

Tancik M, Srinivasan P, Mildenhall B, Fridovich-Keil S, Raghavan N, Singhal U, Ramamoorthi R, Barron J, Ng R. Fourier features let networks learn high frequency functions in low dimensional domains. Advances in neural information processing systems. 2020;33:7537-47.
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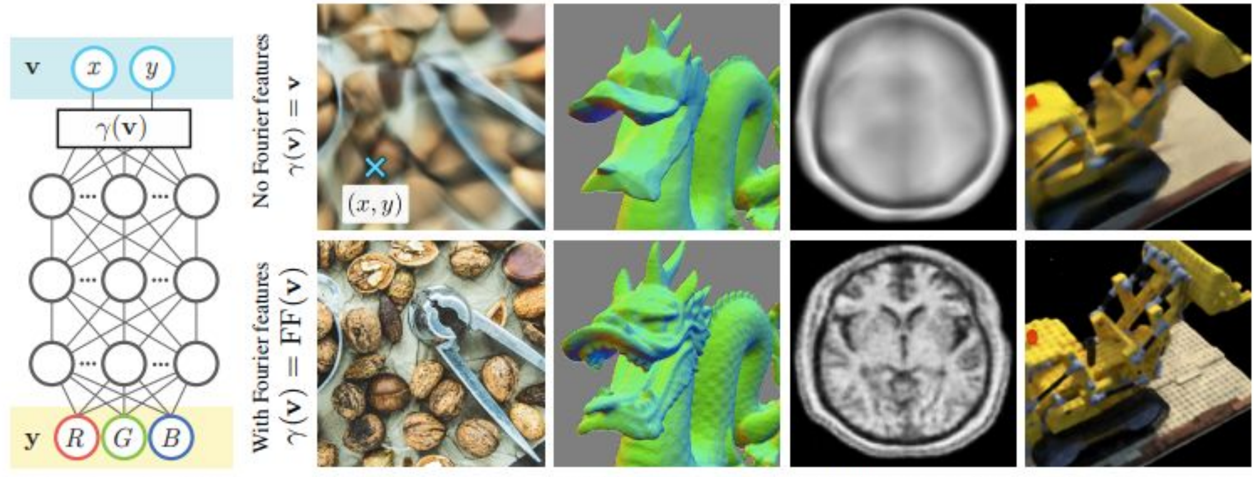
Implicit Neural Representations

Fourier Features

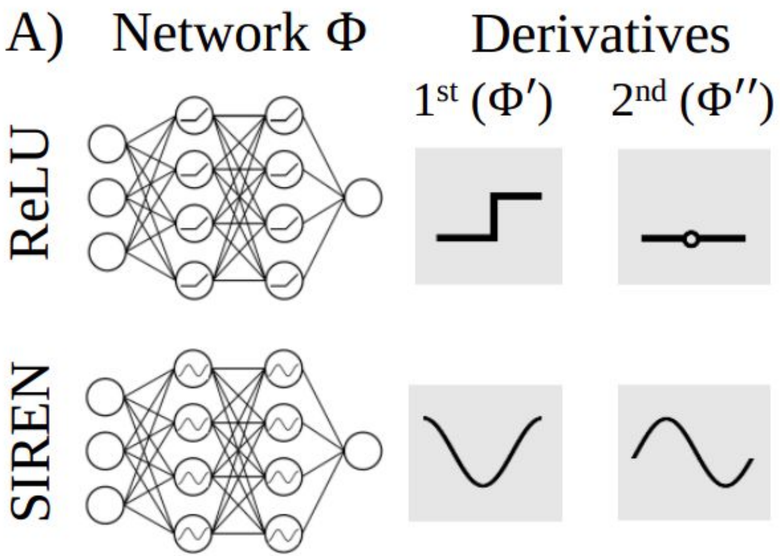


(a) Coordinate-based MLP
(b) Image regression $(x,y) \rightarrow$ RGB
(c) 3D shape regression $(x,y,z) \rightarrow$ occupancy
(d) MRI reconstruction $(x,y,z) \rightarrow$ density
(e) Inverse rendering $(x,y,z) \rightarrow$ RGB, density

Tancik, Matthew, et al. "Fourier features let networks learn high frequency functions in low dimensional domains." *NIPS 2020*

Mitigating the low-frequency bias using SIREN

Sinusoid Activation Functions (SIREN):



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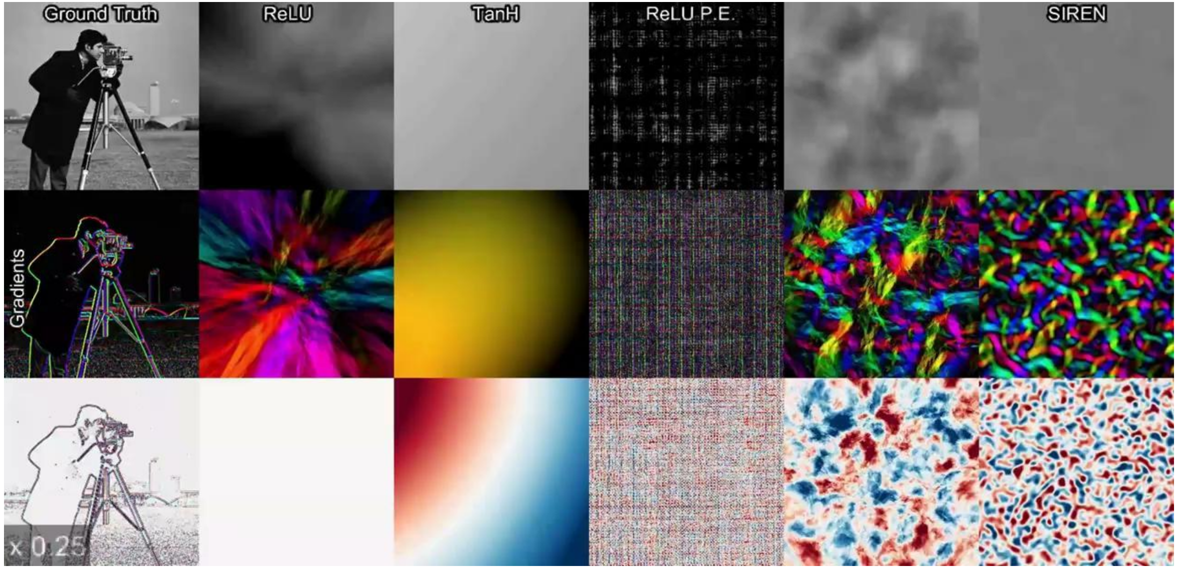
Implicit Neural Representations

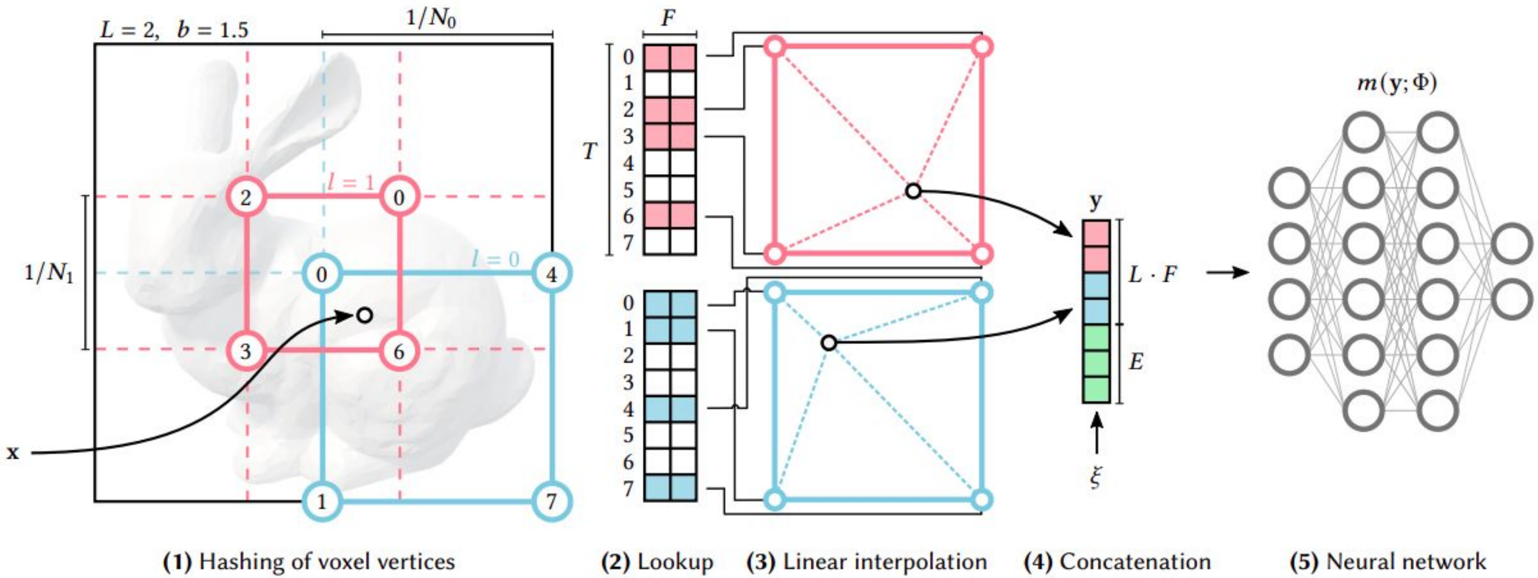
Sinusoidal **Re**presentation **N**etworks (SIREN):



Sitzmann et al. "Implicit Neural Representations with Periodic Activation Functions." *NIPS 2020,*
Video Source: https://www.vincentsitzmann.com/siren/

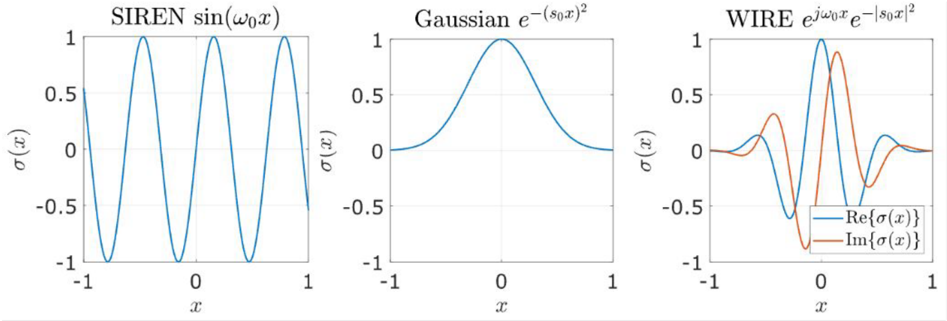
Are the more ?

Hash Grid Encoding

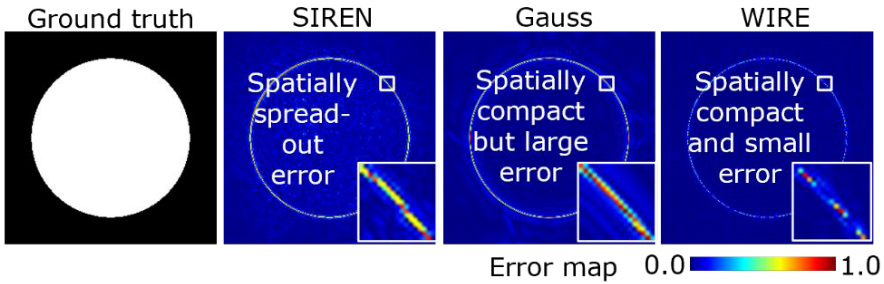


Müller T, Evans A, Schied C, Keller A. Instant neural graphics primitives with a multiresolution hash encoding. ACM transactions on graphics (TOG). 2022 Jul 22;41(4):1-5.

Other activation function



(a) Nonlinearity for Implicit Neural Representations

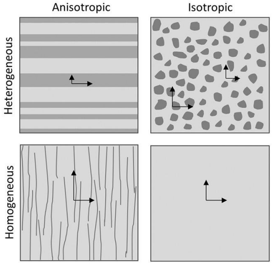


(b) Approximation accuracy with various nonlinearities

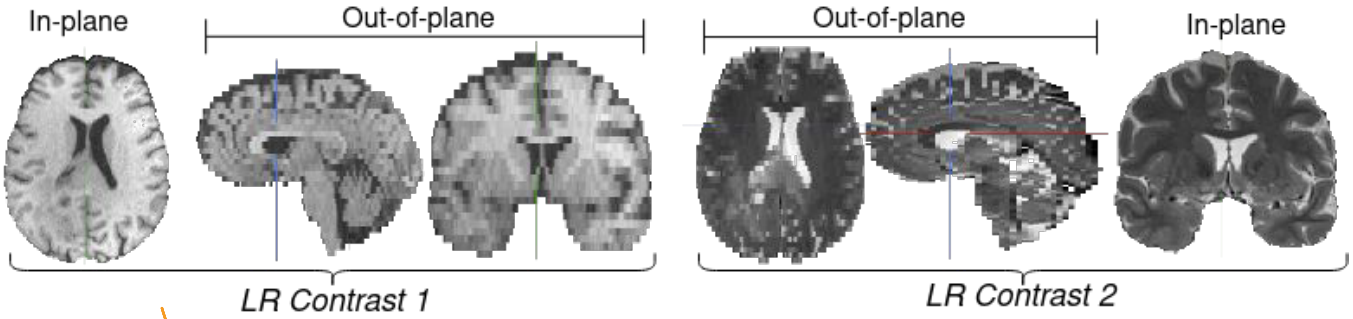
Müller T, Evans A, Schied C, Keller A. Instant neural graphics primitives with a multiresolution hash encoding. ACM transactions on graphics (TOG). 2022 Jul 22;41(4):1-5.

Now we have a compact neural representation of data - why is this helpful?

Motivation



Multi-parametric MRIs are ubiquitous in *clinical protocol* and *retrospective cohorts*

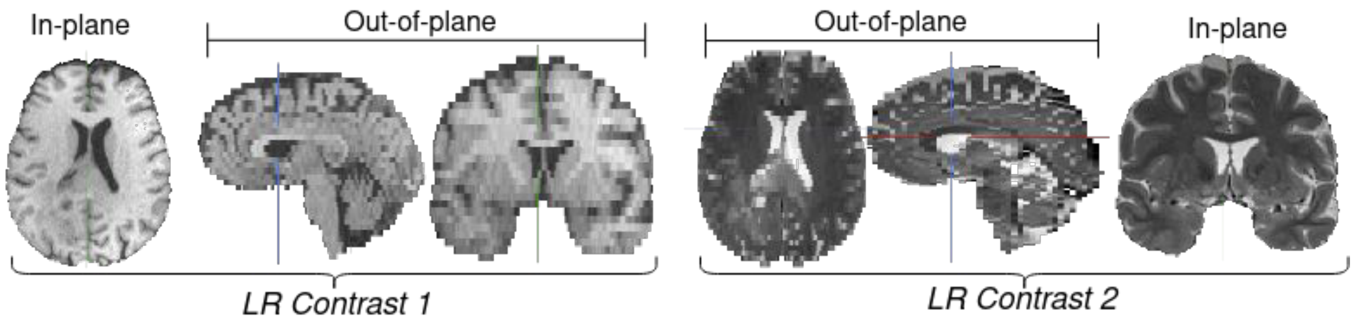


! Most of the scans are obtained using anisotropic 2D protocol → **excellent in-plane** but **poor through plane** details

In-planes are discrete
making challenge in construction
for out-of-planes

Motivation

Multi-parametric MRIs are ubiquitous *in clinical protocol* and *retrospective cohorts*



! Most of the scans are obtained using anisotropic 2D protocol → **excellent in-plane** but **poor through plane** details

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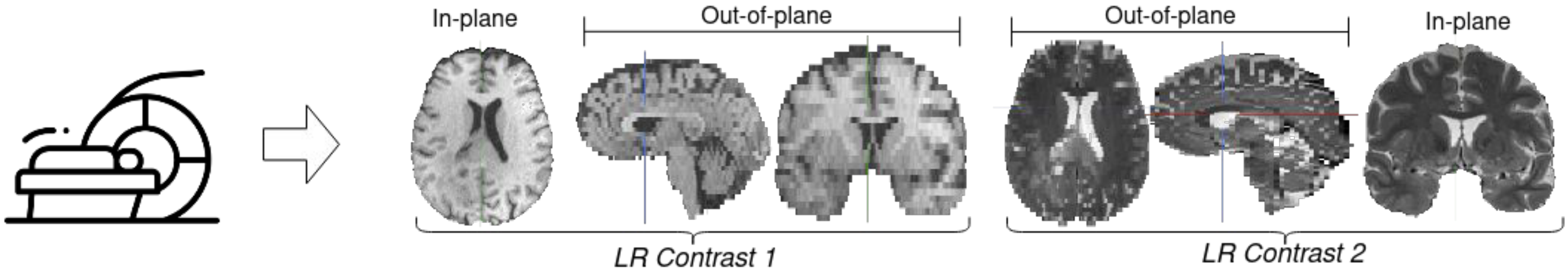
Motivation

Multi-parametric MRIs are ubiquitous in *clinical protocol and retrospective cohorts*



! Most of the scans are obtained using anisotropic 2D protocol → **excellent in-plane** but **poor through plane** details

! Different contrasts are often in complementary views → Radiologist has to bear **higher cognitive** burden

! Most volumetric downstream task rely on ISO 3D scans → **Decreased performance**, e.g., radiomics, lesion volume

Problem Formulation

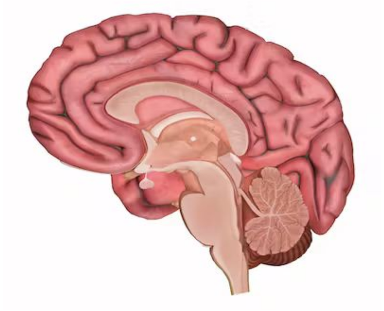
position functions

$$q : \Omega \to \mathbb{A}$$

Anatomical Function

$$\Omega = \{(x, y, z)\}$$

Anatomical Space



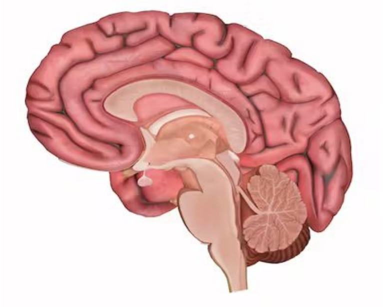
$$\mathbb{A}$$

Anatomical Properties

Problem Formulation

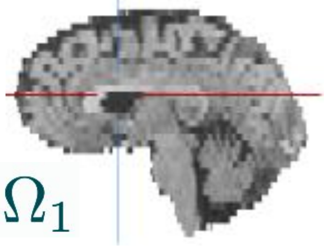
$q : \Omega \rightarrow \mathbb{A}$

Anatomical Function



pixel

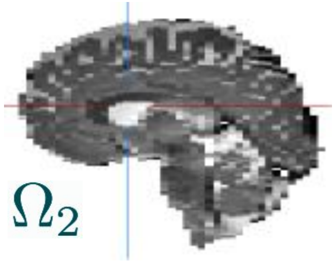
$g_1 : \mathbb{A} \rightarrow \mathbb{R}$



Ω_1

$I_1 = \{g_1(q(\boldsymbol{x})) = f_1(\boldsymbol{x}); \forall \boldsymbol{x} \in \Omega_1\}$

$g_2 : \mathbb{A} \rightarrow \mathbb{R}$



Ω_2

$\Omega = \{(x, y, z)\}$

Anatomical Space

$\mathbb{A}$

Anatomical Properties

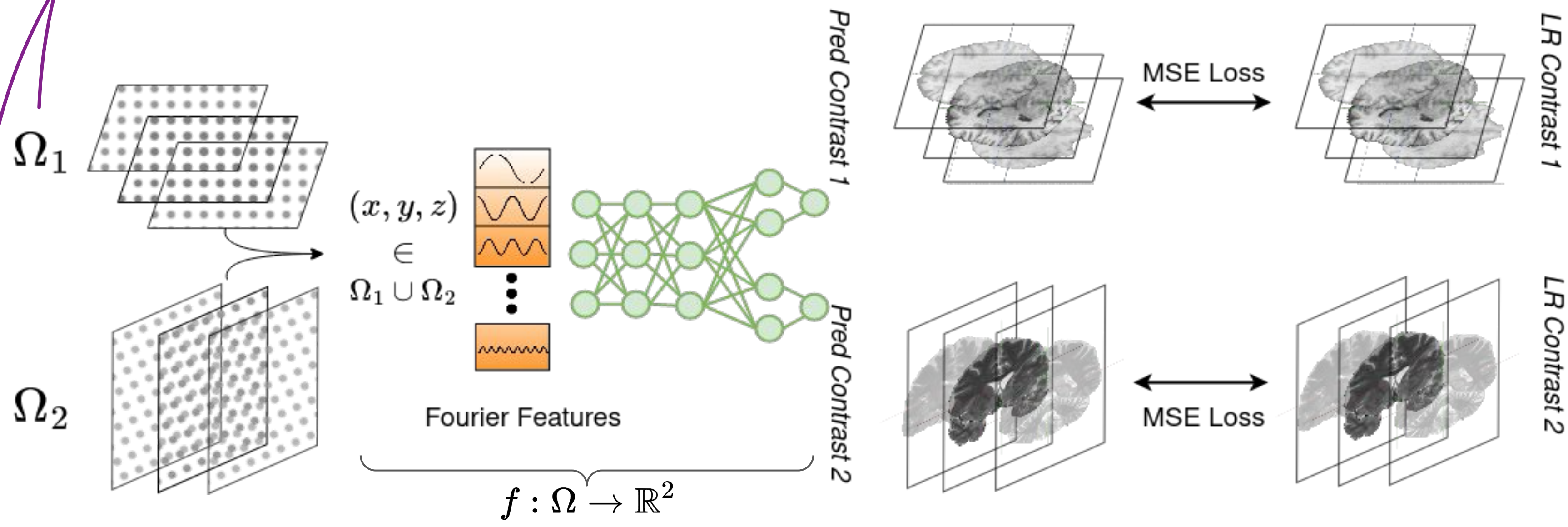
g_1, g_2

Contrast Functions

$I_2 = \{g_2(q(\boldsymbol{x})) = f_2(\boldsymbol{x}); \forall \boldsymbol{x} \in \Omega_2\}$

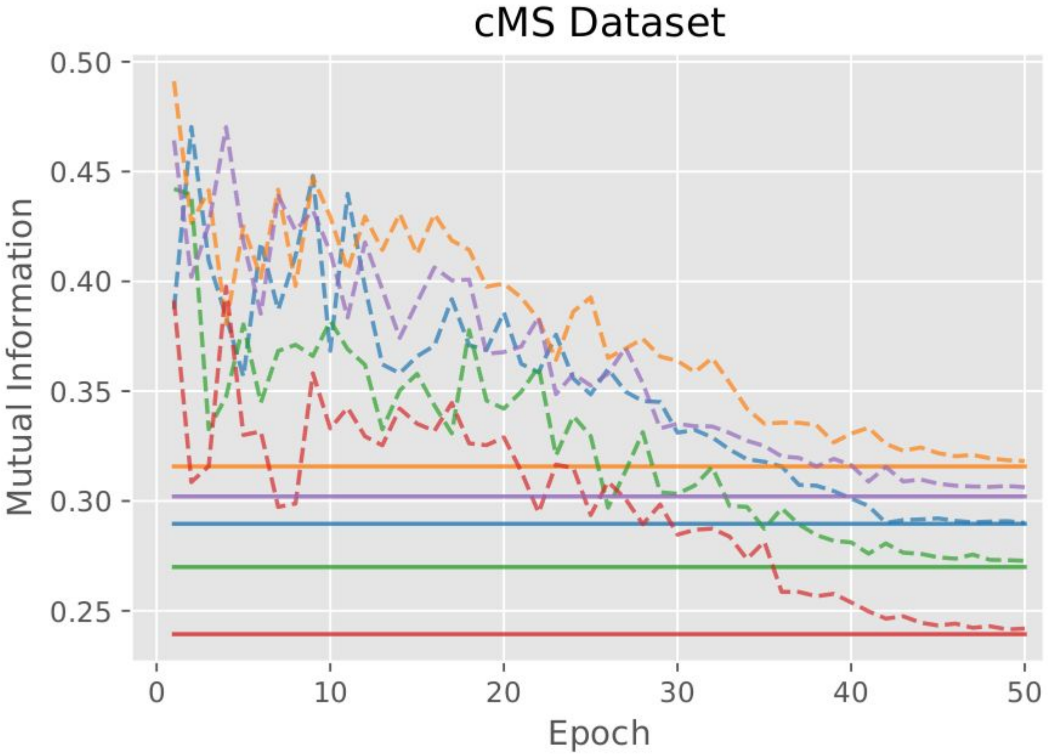
Subject-specific Model Training

2 perspectives

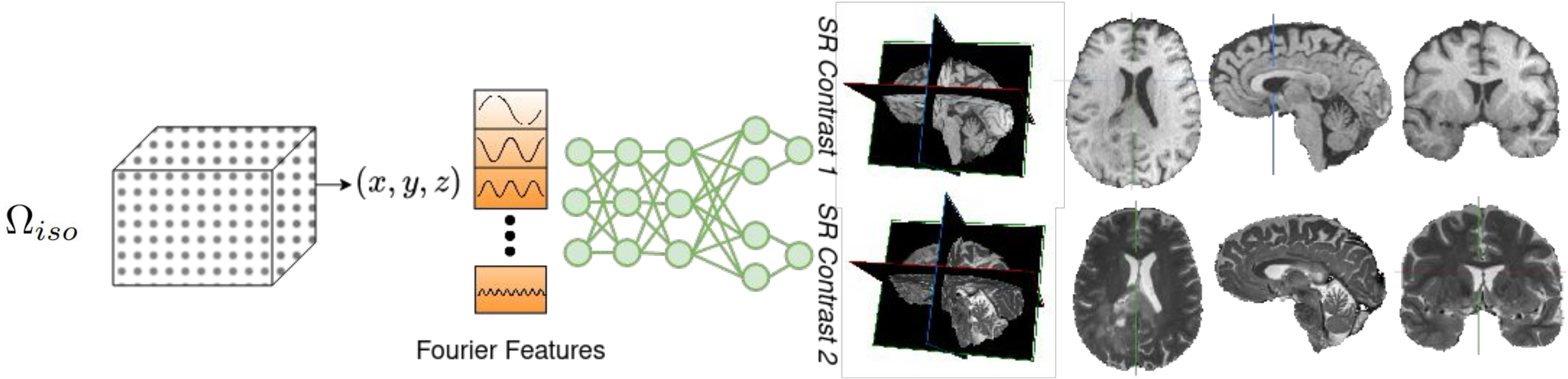


$$\mathcal{L}_{MSE} = \alpha \sum_{\boldsymbol{x} \in \Omega_1} \left(\hat{I}_1(\boldsymbol{x}) - I_1(\boldsymbol{x}) \right)^2 + \beta \sum_{\boldsymbol{x} \in \Omega_2} \left(\hat{I}_2(\boldsymbol{x}) - I_2(\boldsymbol{x}) \right)^2$$

Information Transfer during Training (Mutual Information)

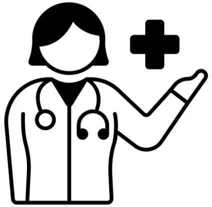


Subject-specific Inference



Inference: Query all coordinates in Ω_{iso} to retrieve super-resolved MRI of both contrasts.

Advantages of Multi-Contrast INRs

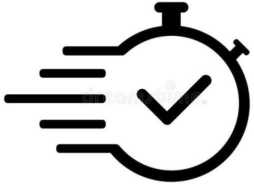


Clinically appealing

- multi-contrast super-resolution without high-resolution training data
- applicable across different pair of contrasts and views

Anatomically faithful

- Information gain is validated by mutual information
- Subject specific and hence not prone to cohort driven hallucination

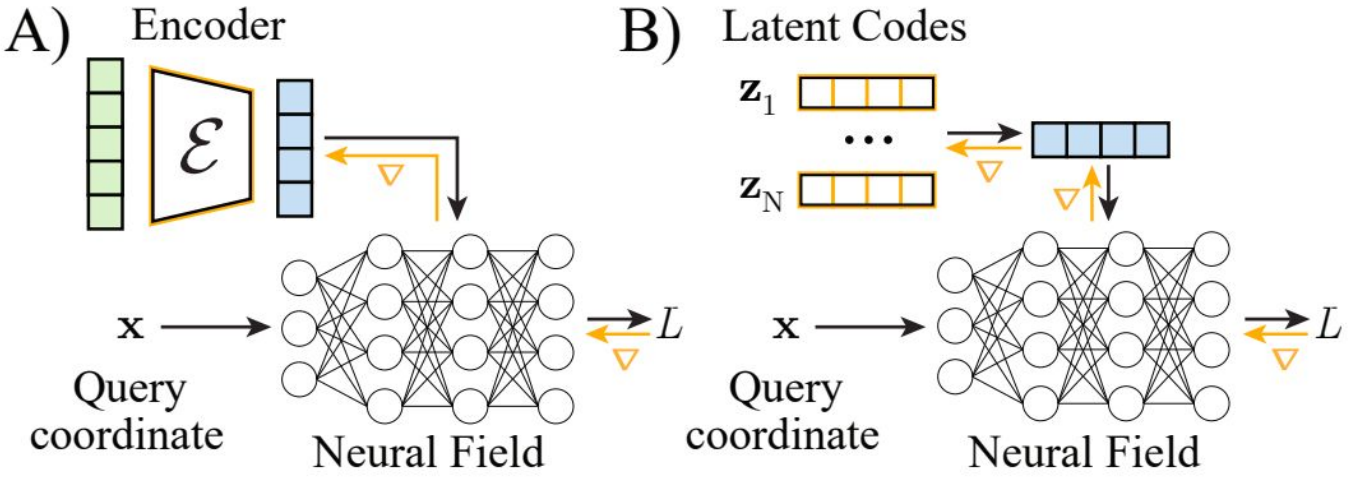


Accelerated computation

- Each super resolution takes few minutes in commodity GPU

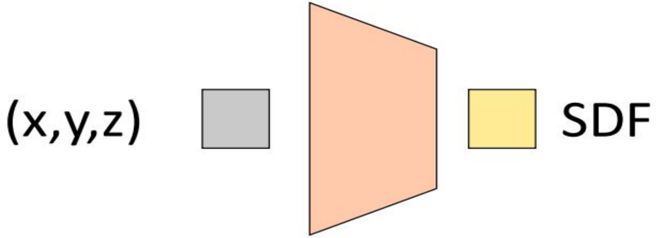
Subject-specific models are meaningful
- but can we also leverage the bias learned across patients in a dataset?

Conditioning INRs



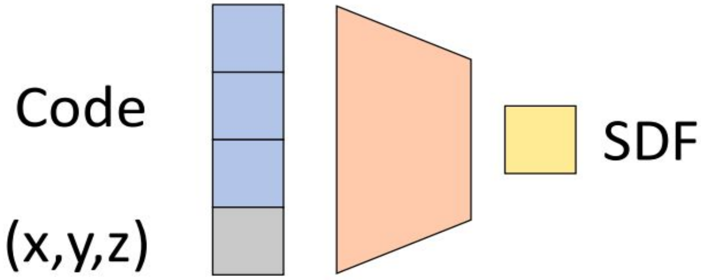
Xie Y, Takikawa T, Saito S, Litany O, Yan S, Khan N, Tombari F, Tompkin J, Sitzmann V, Sridhar S. Neural fields in visual computing and beyond. In: Computer Graphics Forum 2022 May (Vol. 41, No. 2, pp. 641-676).

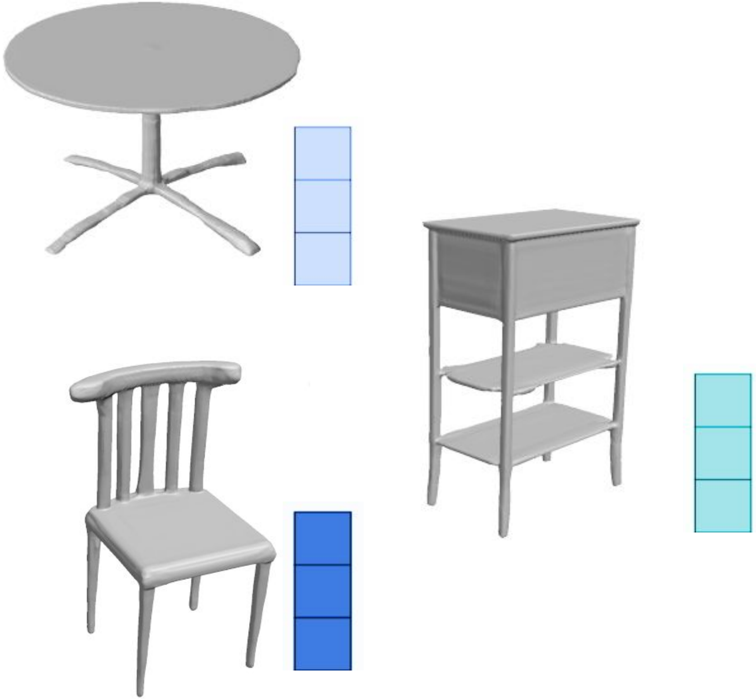
Conditioning INRs



Park JJ, Florence P, Straub J, Newcombe R, Lovegrove S. DeepSDF, CVPR 2019

Conditioning INRs





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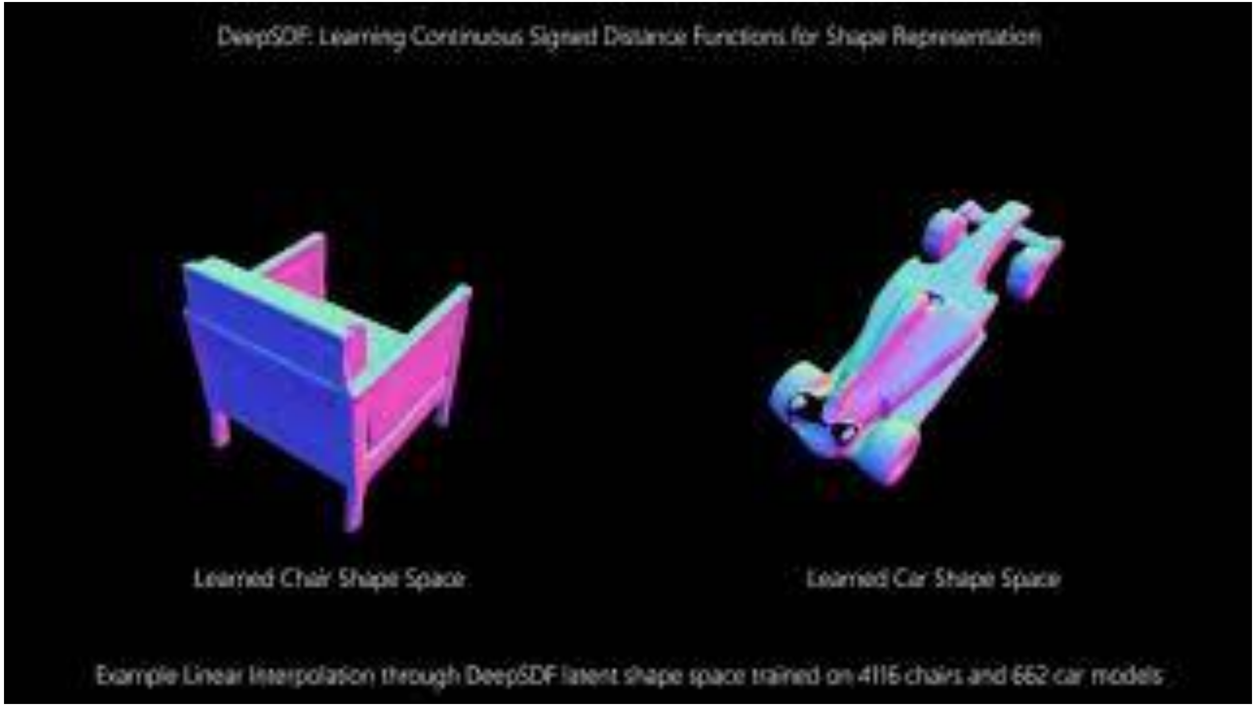
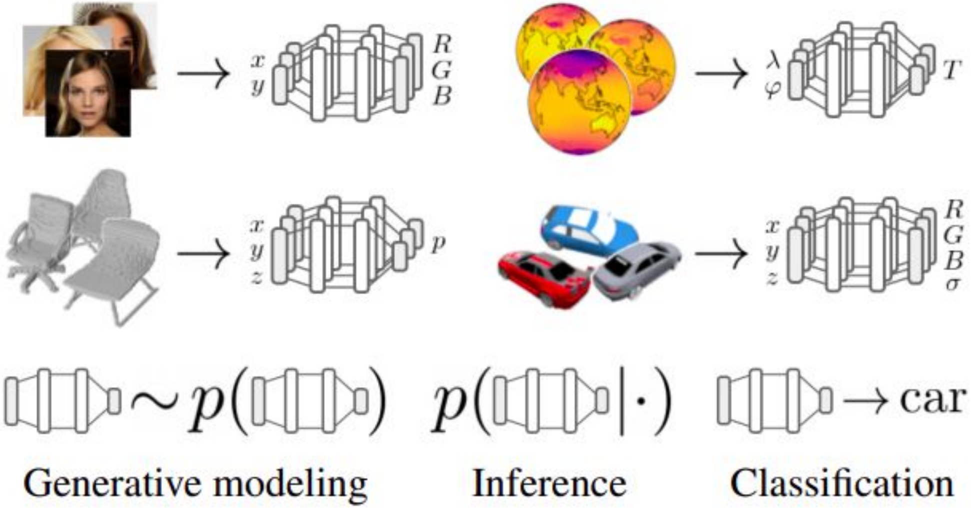
Conditioning INRs



Park JJ, Florence P, Straub J, Newcombe R, Lovegrove S. DeepSDF, CVPR 2019

Conditioning, meta-learning and compressing neural fields



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